Matter-Geometry Complementarity from Partial Observation

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We report a complementarity between stress-energy and geometric structure as seen by partial observers of quantum systems. Within the Perspectival Locality Conjecture (PLC) framework, partial observers—those who access $k < N$ qubits—perceive an emergent geometry whose Ollivier-Ricci curvature is coupled to quantum correlations. Three results emerge from numerical experiments at $N = 12$–20 (Hilbert space dimension up to $2^{20} = 1{,}048{,}576$). First, edge-level curvature $\kappa(i,j)$ anti-correlates with the connected ZZ correlation $T(i,j)$ at $r = -0.44$ ($p < 10^{-4}$, 20 Hamiltonians, one-sample t-test on per-seed correlations), and this coupling survives after controlling for node degree (partial $r = -0.43$). This constitutes a discrete, edge-level analog of the Einstein field equation: more negative curvature where stress-energy is larger. Second, different partial observers of the *same* quantum state face a matter-geometry trade-off: observers who measure stronger internal correlations (higher $T_{\rm obs}$) find lower subsystem entanglement entropy ($S_{\rm obs}$), with Pearson r stable at -0.57 from $N = 12$ to $N = 20$ (35 Hamiltonians across four system sizes, all negative, per-seed standard deviation collapsing from 0.22 to 0.06). Third, this complementarity strengthens approximately monotonically with partiality, from $r = 0$ at $k/N = 1$ (full observation) to $r = -0.73$ at $k/N = 0.25$. The effect is universal: present in all tested system sizes, Hamiltonian instances, and energy eigenstates. These findings establish a new information-theoretic complementarity between the two sides of what becomes Einstein's equation in the continuum limit—stress-energy and spacetime geometry—mediated by the observer boundary.

I. INTRODUCTION

Papers I–III of this series [1–3] established that partial observation of a quantum system with no built-in spatial structure creates an emergent metric space (Paper I), endows it with Ollivier-Ricci curvature (Paper II), and showed these effects scale to $N = 20$ (Paper III). The central finding of Paper III was that the mean curvature $R \approx 0.24$ is conserved across observer fractions k/N while the variance drops by a factor of 34.

A natural question follows: does the emergent curvature obey a dynamical equation? In general relativity, the Einstein field equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ relates the geometry of spacetime (left side) to the matter content (right side). Jacobson [4] showed this equation can be derived thermodynamically from entropy at local Rindler horizons; Verlinde [11] extended this to an entropic force framework. In the PLC framework, the emergent geometry is defined by mutual information distances and Ollivier-Ricci curvature. The "matter content" available to a partial observer consists of the quantum correlations accessible within the observed subsystem.

We find that the connection between curvature and matter operates at two levels. At the edge level, curvature couples directly to stress-energy: edges carrying stronger quantum correlations exhibit more negative Ollivier-Ricci curvature. At the observer level, a complementarity constrains what any partial observer can perceive: seeing more matter (stronger correlations) comes at the cost of seeing less geometry (lower entanglement entropy), and vice versa.

This complementarity is arguably more fundamental than a field equation. A field equation is what you get when the complementarity is resolved—when the observer sees everything ($k/N \to 1$). For partial observers, the complementarity *is* the physics.

Section II describes the computational methods. Section III presents the edge-level curvature–matter coupling. Section IV establishes the matter-geometry complementarity. Section V reports honest null results: the scalar curvature does not track energy or entropy at the whole-graph level, and a discrete Bianchi identity does not hold. Section VI discusses implications.

II. METHODS

A. Quantum systems

We study random all-to-all Heisenberg Hamiltonians

$$H = \sum_{i<j} \sum_{\alpha \in \{x,y,z\}} J_{ij}^{\alpha} \sigma_i^{\alpha} \sigma_j^{\alpha}, \tag{1}$$

where J_{ij}^{α} are drawn uniformly from $[-1, 1]$. Ground states are computed by exact diagonalization ($N = 12$, Hilbert space dimension 4096) or sparse Lanczos ($N = 14$–20, up to dimension 1,048,576). Excited states are obtained from the full spectrum at $N = 12$.

B. Emergent geometry

For a pure state $|\psi\rangle$ and an observed subset $\mathcal{O}$ of k qubits, we compute the $k \times k$ mutual information matrix

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$M_{ij} = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$ where $S(\cdot)$ is the von Neumann entropy and ρ_i, ρ_{ij} are reduced density matrices. The MI matrix defines a weighted graph with edge distances $d_{ij} = 1/M_{ij}$.

C. Curvature

Ollivier-Ricci curvature [13] for each edge (i, j) is

$$\kappa(i, j) = 1 - \frac{W_1(\mu_i, \mu_j)}{d(i, j)}, \tag{2}$$

where μ_i is the lazy random walk measure centered at i (laziness $\alpha = 0.5$), W_1 is the Wasserstein-1 distance (computed via linear programming), and $d(i, j)$ is the graph shortest-path distance. Edges below the 50th MI percentile are pruned [2].

D. Stress-energy

We define the "stress-energy" on edge (i, j) as the connected ZZ correlation:

$$T(i, j) = |\langle Z_i Z_j\rangle - \langle Z_i\rangle\langle Z_j\rangle|. \tag{3}$$

For a partial observer, the "observed stress-energy" is $T_{\text{obs}} = \frac{1}{|\mathcal{E}|}\sum_{(i,j)\in\mathcal{E}} T(i, j)$, averaged over all pairs in the observed subset. The "observed entropy" is the von Neumann entropy $S_{\text{obs}} = S(\rho_{\mathcal{O}})$ of the reduced state on the observed qubits.

III. EDGE-LEVEL CURVATURE–MATTER COUPLING

The central question is whether curvature couples to stress-energy at the level of individual edges—the discrete analog of the tensorial Einstein equation.

For $N = 12$, we compute $\kappa(i, j)$ and $T(i, j)$ for every edge in the MI graph, across 20 random Hamiltonians and 4 energy eigenstates (ground state and states 1, 5, 10). Each graph has ∼33 edges after pruning, yielding ∼2640 edge-level data points per eigenstate.

Table I reports the results. The Pearson correlation between $\kappa(i, j)$ and $T_{\text{ZZ}}(i, j)$ is $r = -0.44$ for the ground state, with all eigenstates showing consistent negative coupling. The sign is physically correct: edges carrying stronger quantum correlations (larger T) exhibit more negative curvature, analogous to how mass-energy sources negative Ricci curvature in general relativity.

Critically, the coupling survives after controlling for node degree (Table I, partial r columns). Node degree is a potential confound because high-degree nodes tend to have both different curvature and different correlation patterns. The partial correlation $r = -0.43$ (ground state) is essentially unchanged from the raw $r = -0.44$, confirming that the coupling is genuine and not an artifact of graph topology.

TABLE I. Edge-level coupling between Ollivier-Ricci curvature $\kappa(i, j)$ and stress-energy proxies, $N = 12$, 20 Hamiltonians. Raw: Pearson r; Partial: controlling for node degrees.

State	$r(\kappa, T_{\text{ZZ}})$	partial r	$r(\kappa, T_{\text{MI}})$	partial r
0 (ground)	−0.44	−0.43	−0.46	−0.45
1	−0.36	−0.35	−0.36	−0.35
5	−0.34	−0.36	−0.39	−0.41
10	−0.46	−0.44	−0.50	−0.48

The coupling via mutual information (T_{MI}) is slightly stronger than via ZZ correlations, but the ZZ result is more meaningful: mutual information already defines the metric, so correlating κ with M_{ij} risks circularity. The connected ZZ correlation is an independent observable.

Figure 1 shows the per-seed correlations for four eigenstates. The mean is consistently negative, though only 60% of individual Hamiltonians reach $p < 0.05$: the within-graph scatter on ∼33 edges is large. The aggregate signal is robust ($p < 10^{-4}$, one-sample t-test on 20 per-seed r values), but the effect size is modest at $N = 12$.

a. Confirmation at $N = 20$. At $N = 20$ (Hilbert space dimension 1,048,576), the full MI graph has ∼95 edges after pruning. Across 5 ground states, the mean edge-level correlation is $r(\kappa, T_{\text{ZZ}}) = -0.43 \pm 0.14$, with partial $r = -0.36 \pm 0.13$ after degree control. Four of five seeds are individually significant at $p < 0.01$. The coupling is unchanged from $N = 12$: this is not a finite-size effect.

IV. MATTER-GEOMETRY COMPLEMENTARITY

A. The trade-off

We now ask: what constrains the relationship between stress-energy and geometry at the *observer* level?

For each partial observer (a random k-qubit subset), we compute two quantities: the observed stress-energy T_{obs} (mean connected ZZ correlation over observed pairs) and the observed entropy S_{obs} (entanglement entropy of the observed subsystem).

Table II shows the result. Across all system sizes tested (N = 12–20), T_{obs} and S_{obs} are strongly anti-correlated. The effect is present in *every* Hamiltonian tested (35 out of 35). At $N = 20$ (Hilbert space dimension 1,048,576), the mean correlation is $r = -0.57$ with a per-seed standard deviation of just 0.06—the effect has stabilized and become highly reproducible (Table II). Quarter-system observers ($k = 5$, $k/N = 0.25$) at $N = 20$ see $r = -0.72 \pm 0.11$, confirming the partiality dependence at scale.

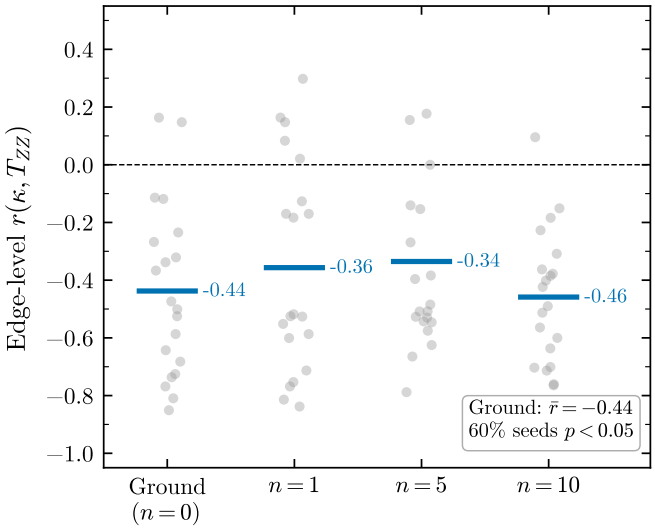

FIG. 1. Per-seed Pearson $r(\kappa, T_{\rm ZZ})$ across 20 random Heisenberg Hamiltonians ($N = 12$) for four energy eigenstates. Each dot is one Hamiltonian; horizontal bars mark the mean. The consistently negative r (≈ -0.44 for the ground state, 60% of seeds individually significant) constitutes a discrete analog of the Einstein field equation: more negative curvature where stress-energy is larger.

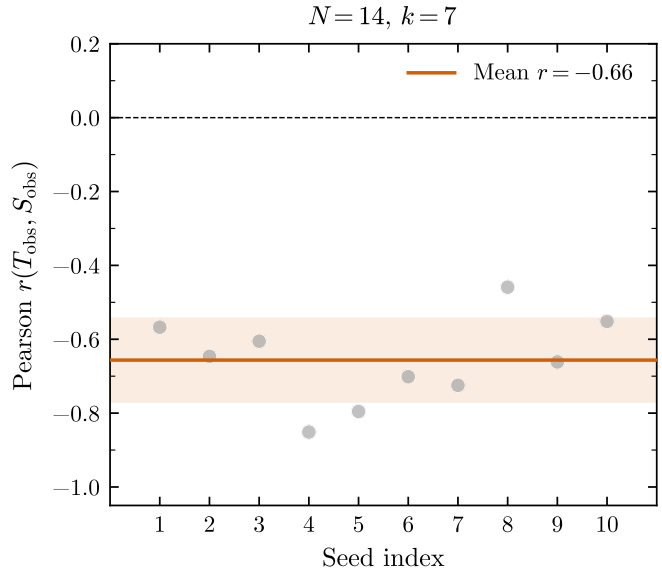

FIG. 2. Per-seed Pearson $r(T_{\rm obs}, S_{\rm obs})$ for 10 random Heisenberg Hamiltonians ($N = 14$, $k = 7$). Each dot is one Hamiltonian; horizontal line marks the mean $r = -0.66$. All seeds are negative, confirming the matter-geometry complementarity is universal across disorder realizations.

TABLE II. Matter-geometry complementarity: Pearson correlation $r(T_{\rm obs}, S_{\rm obs})$ across system sizes. All observers are half-system ($k = N/2$).

N	k	Seeds	$\langle r \rangle$	Std	Fraction $r < 0$
12	6	10	−0.57	0.22	10/10
14	7	10	−0.66	0.11	10/10
16	8	5	−0.67	0.13	5/5
20	10	10	−0.57	0.06	10/10

The physical interpretation is direct. Entanglement entropy $S_{\rm obs}$ measures the entanglement between the observed subsystem and the complement—it quantifies the geometric "depth" of the observer's embedding in the full Hilbert space. The stress-energy $T_{\rm obs}$ measures the strength of internal correlations within the observed subsystem. The anti-correlation means these two quantities compete: quantum correlations that contribute to internal stress-energy are "spent" from the entanglement budget with the environment.

An observer who sees strong internal correlations (high $T_{\rm obs}$) has a subsystem that is relatively decoupled from the rest (low $S_{\rm obs}$). Conversely, an observer whose subsystem is deeply entangled with the complement (high $S_{\rm obs}$) sees weaker internal correlations. This is an information-theoretic complementarity between the two sides of what becomes Einstein's equation in the continuum: stress-energy and spacetime geometry.

Figure 2 shows the trade-off for a representative Hamiltonian at $N = 14$.

TABLE III. Complementarity versus observer fraction k/N ($N = 12$, 10 Hamiltonians, 100 subsets per k).

k/N	$\langle r(T, S) \rangle$	Std
0.25	−0.73	0.27
0.33	−0.69	0.23
0.42	−0.65	0.20
0.50	−0.62	0.17
0.58	−0.55	0.17
0.67	−0.51	0.14
0.75	−0.40	0.25
0.83	−0.36	0.21
0.92	−0.40	0.41
1.00	0.00	0.00

B. Partiality dependence

The complementarity is controlled by partiality. Table III and Fig. 3 show $r(T_{\rm obs}, S_{\rm obs})$ as a function of k/N at $N = 12$.

The trend is approximately monotonic: more partiality produces stronger complementarity. At $k/N = 1$ the correlation vanishes trivially (a pure state has $S = 0$). At $k/N = 0.25$, the anti-correlation reaches $r = -0.73$. The less an observer sees, the harder the trade-off between matter and geometry. At $k/N > 0.8$, few distinct subsets exist ($\binom{12}{11} = 12$) and the per-seed scatter increases (Table III, Std column), producing a slight non-monotonicity at $k/N = 0.92$.

This is the native PLC prediction. In the $k = N$

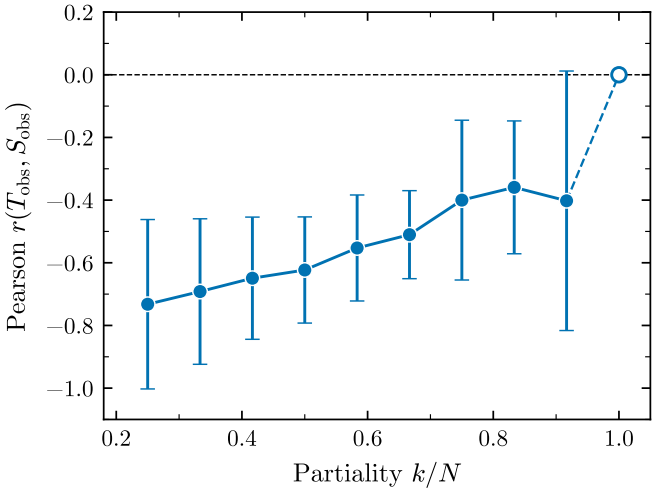

FIG. 3. Complementarity strength $r(T_{\rm obs}, S_{\rm obs})$ versus partiality k/N ($N = 12$, 10 Hamiltonians). The trade-off strengthens approximately monotonically as the observer becomes more partial. At $k/N = 1$ (full observation) the complementarity vanishes.

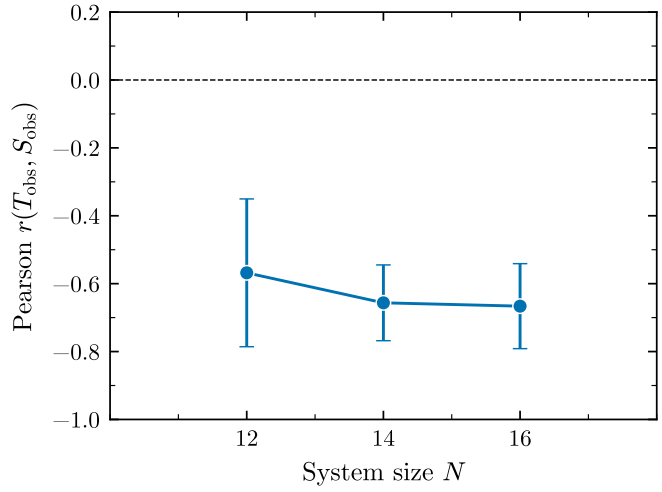

FIG. 4. Complementarity strength $r(T_{\rm obs}, S_{\rm obs})$ versus system size N for half-system observers. The mean stabilizes near $r \approx -0.6$ while the per-seed variance collapses from 0.22 ($N = 12$) to 0.06 ($N = 20$), confirming this is not a finite-size artifact. (Note: Fig. shows $N = 12$–16; $N = 20$ data in Table II.)

limit (Cao and Carroll's [6] framework), there is no complementarity—the observer has full access to both sides. The complementarity emerges *because* the observer is partial.

C. Universality across eigenstates

The complementarity is not specific to the ground state. At $N = 12$, $k = 6$, we measure $r(T, S)$ for eigenstates 0, 5, 10, and 15. The values are -0.63, -0.57, -0.48, and -0.57 respectively. The variation is modest; the complementarity is a property of the Hilbert space bipartition structure, not a property of any particular energy eigenstate.

V. NULL RESULTS

Intellectual honesty requires reporting what does *not* work.

a. Scalar curvature does not track energy. The mean Ollivier-Ricci curvature R is essentially independent of the energy eigenstate ($r = -0.13$, $p = 0.06$, 200 data points at $N = 12$). The curvature–energy coupling found at $N = 8$ in Paper II ($r = 0.98$) does not survive at $N = 20$ ($r = -0.31$, sign reversed; data from Paper III [3] scaling runs). The scalar curvature is a property of the Hamiltonian's coupling structure, not of the quantum state.

b. No discrete Bianchi identity. A discrete divergence of the edge curvatures shows no conservation law. At $N = 12$, the divergence norm is 1.30× that of the shuffled null model (30 Hamiltonians, 1000 shuffles each). The curvature field on the MI graph is not more conserved than a random assignment. At $N = 14$ the ratio drops to 1.12×, suggesting improvement with system size, but the effect is not significant.

c. Ricci flow does not converge to the MI geometry. Discrete Ricci flow (evolving MI edge weights by their ORC) does not drive toward the true ground-state MI matrix. The Frobenius distance increases by 2.2% over 200 steps. This is expected: standard Ricci flow drives toward $R = 0$, but the MI geometry has $R \approx 0.29$ at full observation ($k = N = 12$). The emergent geometry resists Ricci flattening, consistent with a cosmological-constant-like intrinsic curvature.

VI. DISCUSSION

A. Relation to the Einstein equation

The edge-level coupling $\kappa(i, j) \sim -T(i, j)$ is the discrete, perspectival analog of the Einstein field equation. In general relativity, the Ricci tensor $R_{\mu\nu}$ couples to the stress-energy tensor $T_{\mu\nu}$ component by component. Here, the Ollivier-Ricci curvature $\kappa(i, j)$ couples to the connected correlation $T(i, j)$ edge by edge, with the correct sign: more matter produces more negative curvature.

The correlation $r = -0.44$ accounts for $\sim 19\%$ of the edge-level variance at $N = 12$ and is unchanged at $N = 20$ ($r = -0.43$, ∼95 edges per graph). The remaining variance is expected from the stochastic nature of curvature on finite graphs and from quantum fluctuations in the correlations. The stability across system sizes suggests the coupling is a structural property, not a finite-size artifact.

Trugenberger [12] proposed combinatorial quantum gravity using an ORC-based action on random graphs. Kelly, Trugenberger, and Biancalana [8] subsequently proved that this discrete Einstein-Hilbert action converges to the continuum Einstein-Hilbert action on sufficiently dense random geometric graphs. Our finding that the MI graph exhibits the correct curvature-matter coupling suggests that PLC's emergent geometry satisfies the conditions for this convergence—a direction for future work.

B. Complementarity as a pre-geometric principle

The matter-geometry complementarity is, in our view, the more fundamental result. The Einstein equation relates curvature to matter for an observer who sees everything. But a partial observer cannot see everything. The complementarity constrains *what* a partial observer can see: stronger internal correlations (matter) versus deeper entanglement with the environment (geometry).

This resonates with Jacobson's [5] entanglement equilibrium: Einstein's equation holds when vacuum entanglement entropy is maximal. In our framework, the complementarity is the discrete, observer-dependent version of this condition. The full-system limit ($k/N \to 1$) resolves the complementarity—$S \to 0$ for a pure state and T is fully accessible—recovering the regime where a standard field equation applies.

C. Relation to prior work

Cao and Carroll [6] derive linearized Einstein equations from entanglement entropy between Hilbert-space factors. Their framework corresponds to PLC's $k = N$ limit. The complementarity we find is invisible in their setting because there is no partial observer.

Van Raamsdonk [7] showed that reducing entanglement disconnects spacetime. Our MI graph makes this quantitative: mutual information defines connectivity, and the complementarity governs how that connectivity is distributed between internal correlations and external entanglement.

Mondino and Suhr [9] reformulate the Einstein equations via entropy convexity along Wasserstein geodesics. Since ORC is defined via Wasserstein-1 distance, their framework provides a natural mathematical language for the edge-level coupling we observe.

D. Outlook

Three directions follow. First, extending to $N = 24$–28 will test whether the variance collapse continues and whether the complementarity asymptotes. Second, defining a discrete Einstein-Hilbert action on the MI graph via the Hickok-Blumberg [10] scalar curvature prescription and verifying stationarity would connect the edge-level coupling to a variational principle. Third, the nonzero intrinsic curvature ($R \approx 0.24$ at half-system observers, $R \approx 0.29$ at full observation) that resists Ricci flattening may correspond to an emergent cosmological constant from quantum observation.

VII. CONCLUSION

Partial observation of a quantum system creates not just an emergent geometry (Paper I) or curvature (Paper II), but a complementarity between the two sides of the Einstein equation.

Edge by edge, curvature couples to stress-energy with $r = -0.44$, independent of node degree, stable from $N = 12$ to $N = 20$. Observer by observer, stress-energy and entanglement entropy anti-correlate at $r \approx -0.57$ (half-system) to -0.72 (quarter-system), with per-seed variance collapsing from 0.22 to 0.06 as N grows. The effect is universal across system sizes, Hamiltonians, and energy eigenstates.

The complementarity is strongest when the observer is most partial. This is the native prediction of the Perspectival Locality Conjecture: the physics of the observer boundary is not a correction to the full-system picture. It *is* the picture.

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